

11.4.12.4 Code Review And Quality

2026-06-04

§ 11.4.12.4 — Dead/unwired-code
investigate-before-remove mandate (User mandate,
— —)

Forensic anchor — verbatim user mandate (2026-06-04):

"Before removing any seemingly-dead (zero-importer / unwired) codebase, we MUST investigate via git history where/how it was originally used and how it became dead. Removal is permitted ONLY when we have captured PROOF it is genuinely no longer needed — and that removal MUST be its own separate commit with a proper descriptive message. If there is no such proof, the code MUST be investigated for where/how it should be wired in properly, and any missing or unwired tests MUST be added. We MUST ALWAYS be extra careful with any codebase removal."

The defect class. "This has zero importers / is never called / is unwired, so it is dead — delete it" is a guess (§11.4.6), not a finding. A `git grep` / static-analysis "no references" result proves only *current* non-reference; it does NOT prove the code is genuinely unneeded. Code becomes apparently-dead for several distinct reasons — its call-site was deleted by mistake (a regression: the code SHOULD still be wired in), its wiring was never finished (it is intended but unwired — the fix is to WIRE it, plus its tests), it is reflectively / dynamically / build-tag / plugin / DI-container referenced (the static tool simply cannot see the reference), or it is genuinely obsolete and safe to remove. Deleting it without first distinguishing which case applies risks silently removing a needed-but-temporarily-unwired capability — the same severity class as the §11.4.122 silent-removal defect and a §11.4 PASS-bluff at the maintenance layer.

The mandate. Before removing ANY seemingly-dead codebase element (a zero-importer / never-called / unwired / unreferenced function, method, type, file, module, package, asset, config, or build target), the agent **MUST FIRST** investigate via git history (`git log --follow` , `git log -S<symbol> / -G<regex>` `pickaxe` across all history, `git log --diff-filter` , `blame` on the deletion of the former call-site) to establish, as captured FACT (§11.4.6): (1) WHERE and HOW it was originally used / wired in; (2) WHEN and HOW it became dead — its call-site removed deliberately, removed by mistake (regression), never completed, or made unreachable by a refactor; (3) whether the "no references" result is real OR an artefact of a reference the static tool cannot see (reflection / dynamic dispatch / build-tags / codegen / DI / plugin registry / cross-language FFI / config-driven wiring). The investigation output (the cited commits + the determination) is the captured evidence.

Removal is conditional — proof-of-genuinely-unneeded REQUIRED. Removal is permitted **ONLY** when the investigation produces captured **PROOF** the element is genuinely no longer needed (its purpose is fully superseded, its former caller is intentionally and permanently gone, no hidden reference exists). When that proof exists, the removal **MUST** be its **OWN SEPARATE COMMIT** (not bundled with unrelated work — so it is independently reviewable and revertible, composing with §11.4.84 working-tree quiescence + §11.4.92 multi-pass) carrying a proper descriptive message that cites the git-history investigation evidence (the commits that wired it in, the commit that orphaned it, the determination that it is unneeded) and — where the element is an end-user-facing capability — the §11.4.122 operator-confirmation. The §11.4.90 tracked path applies: classify the tracker item `Obsolete (→ Fixed.md)` with the appropriate reason.

No proof ⇒ wire it in + add the missing tests (do NOT delete). If the investigation does **NOT** yield proof the element is unneeded, the element **MUST NOT** be removed. Instead it **MUST** be investigated for WHERE and HOW it should be wired in properly (restore the deleted call-site if its removal was a regression per §11.4.114; complete the intended wiring if it was never finished), **AND** any missing or unwired tests for it **MUST** be added (the absence of a test is itself part of why it drifted into apparent-deadness — composes §11.4.27 100%-test-type-coverage + §11.4.43 TDD-fix + §11.4.115 RED-baseline). Apparent-deadness with no removal-proof is a wiring/coverage GAP to close, never a delete-on-sight.

Extra-caution default. The agent MUST ALWAYS be extra careful with any codebase removal: when uncertain whether removal-proof is sufficient, default to NOT removing (investigate + wire + test), per §11.4.6 no-guessing + §11.4.101 (an irreversible-ish, possibly-needed removal whose safety cannot be determined from evidence is exactly the class that blocks to the operator rather than being autonomously deleted) + §11.4.122 (removal of a live end-user capability requires operator confirmation). "Probably dead" is never sufficient; the bar is captured proof.

Composition. §11.4.124 composes with §11.4.6 (no-guessing — "zero importers therefore dead" is the exact guess forbidden; the determination must be FACT from git history), §11.4.8 (deep-research — git history IS the primary research corpus for how the code was used + how it died; external research applies when the wiring mechanism is unclear), §11.4.84 / §11.4.92 (the separate-removal-commit discipline — quiescent tree, multi-pass evaluation, independently revertible), §11.4.90 (Obsolete classification is the tracked path for a proven removal), §11.4.101 (autonomous-decision-over-blocking — a removal whose safety cannot be proven blocks rather than proceeds), §11.4.114 (last-known-good-tag isolation — a call-site deleted by mistake is a regression to repair, the same diagnostic the "became dead by mistake" case demands), §11.4.122 (no-silent-removal — §11.4.124 is its complement: §11.4.122 governs removal of KNOWN-shipped capabilities by operator confirmation, §11.4.124 governs removal of SEEMINGLY-dead code by git-history proof; an end-user-facing seemingly-dead element triggers BOTH), §11.4.27 / §11.4.43 / §11.4.115 (the missing-tests obligation when the element is wired in rather than removed).

Enforcement. Propagation gate `CM-COVENANT-114-124-PROPAGATION` enforces the literal anchor `11.4.124` across the consumer fleet (every `CLAUDE.md` / `AGENTS.md` / `QWEN.md`). Recommended gate `CM-DEAD-CODE-INVESTIGATE-BEFORE-REMOVE` (a commit-audit detector: any commit whose diff is a `NET DELETION` of a code element — function/type/file/module/package/asset/build-target — MUST be a removal-only commit whose message cites the git-history investigation evidence, OR the deletion is part of a tracked `obsolete` (→ `Fixed.md`) item per §11.4.90; a net-deletion commit bundled with unrelated changes OR lacking the investigation citation FAILs). Paired §1.1 meta-test mutations: strip the `11.4.124` literal → propagation gate FAILs; introduce a net-

deletion commit with no investigation citation and not part of an Obsolete item →
CM-DEAD-CODE-INVESTIGATE-BEFORE-REMOVE FAILs. (Gate-code implementation
lands as a separate follow-up work item; this anchor defines the contract.)

Classification: universal (§11.4.17) — "investigate git history + capture proof-of-
unnneeded before removing seemingly-dead code, otherwise wire it in and add its
tests; removal is its own descriptive commit" is a platform-neutral maintenance
discipline reusable by ANY codebase; the consuming project supplies its concrete
static-analysis / importer-graph tooling and its hidden-reference mechanisms
(reflection, DI, build-tags, codegen) per §11.4.35.

Canonical authority: this Constitution.md §11.4.124 in the HelixConstitution
submodule. All consuming projects restate + cite via §11.4.35 inheritance.

Non-compliance is a release blocker regardless of context. No escape hatch — no
--zero-importers-means-dead , --delete-unwired-on-sight , --skip-git-
history-investigation , --remove-without-proof ,
--bundle-removal-with-other-work flag exists.

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§ . . — Code-review-agent gate
before pre-build + main build (mandatory multi-
layer review) (User mandate,
- -)

Forensic anchor — verbatim user mandate (2026-06-04):

"After all fixes/changes/implementations are done, BEFORE running pre-build
tests and the main build, dispatch code-review agent(s) that analyze all work
done + all existing data/facts + the existing codebase + current git history to
determine quality, safety, and whether the fixes/changes will REALLY work;
they MUST validate and verify that every test covering the fixes/changes
genuinely validates the work with NO chance of false results or bluff of any
kind. Any finding MUST be fixed, polished, improved, and covered with
additional tests before the build proceeds. Multiple strong layers of checks."

The defect class. Running the pre-build gate sweep and the main (artifact) build directly after a batch of fixes/changes — without an independent review pass first — lets a defective fix, an error-prone change, a bluff test, or a test that does not genuinely validate the work-under-test consume an expensive build cycle and (worse) ship green. Pre-build gates and the build prove the codebase is internally consistent + buildable; they do NOT prove the work is correct, safe, will-really-work for the end user, or that the tests covering it actually catch its negation. That assurance is an independent-review responsibility that MUST run BEFORE the build, not after.

The mandate. After all fixes / changes / implementations in a batch are done, and BEFORE running the pre-build test sweep and the main build (for ANY project), the agent MUST dispatch one or more dedicated code-review agent(s) (subagent-driven by default per §11.4.70 / §11.4.20) that perform a multi-layer review. The review MUST: (1) **analyze ALL work done in the batch** — every fix / change / implementation, its source diff, and its stated intent; (2) **analyze ALL existing data + facts** — captured evidence per §11.4.5 / §11.4.69 / §11.4.107, tracker entries, prior findings, the §11.4.108 runtime-signature registry; (3) **analyze the existing codebase** — the blast radius of each change (§11.4.92 multi-pass), cross-feature interaction, contract integrity of every dependency; (4) **analyze current git history** — what each change touched, how it composes with concurrent / recent work, whether it reproduces a known-broken pattern (§11.4.114 / §11.4.124 git-history discipline); (5) **determine quality, safety, and will-it-REALLY-work** — assess whether each change is robust + not error-prone (will not solve A while creating B), safe (no host-safety / data-safety / security regression), and genuinely delivers the end-user-visible behaviour it claims (§11.4 / §107); (6) **validate + verify the tests covering the work** — confirm every test that covers a fix/change genuinely exercises the work-under-test and catches its negation, with ZERO chance of a false result or bluff of any kind (a test that PASSES on broken-for-the-user work, a metadata-only / config-only / absence-of-error / grep-without-runtime assertion, or a gate whose paired §1.1 mutation does not make it FAIL is a finding the review MUST surface per §11.4.1 / §11.4.6 / §11.4.107 / §1.1).

Findings block the build. Any finding the review surfaces — a defect, an error-prone change, a safety risk, a will-not-really-work conclusion, a bluff/false-result-capable test, a missing-coverage gap — MUST be fixed, polished, improved, and covered with additional tests (four-layer per §11.4.4(b), TDD-RED-first per §11.4.43 / §11.4.115) BEFORE the pre-build sweep and main build proceed. The

review MUST iterate (re-review after each remediation) until it surfaces no remaining blocking findings; only then does the build proceed. The review is itself anti-bluff: its conclusions are captured evidence (§11.4.5 / §11.4.69), and a review that rubber-stamps a defective batch is the same severity class as a PASS-bluff.

Multiple strong layers. The review is one of multiple strong layers of checks, complementing — never replacing — the pre-build gate sweep (§1 four-layer), the §11.4.92 multi-pass change-evaluation, the §11.4.108 four-layer fix-verification, the §11.4.110 pre-build build-readiness verdict + clash detector, and the post-build / runtime-on-clean-target / user-visible layers. §11.4.125 sits at the seam between "all batch work done" and "start the pre-build + build": it is the independent-review gate that the other layers do not provide (the multi-pass per §11.4.92 is self-review by the change author; §11.4.125 adds the structurally-separated reviewer seam per §11.4.70).

Composition. §11.4.125 composes with §11.4 (the end-user-quality covenant the review enforces), §11.4.1 (FAIL-bluffs equally caught), §11.4.4 (four-layer coverage of every remediation the review demands), §11.4.6 (no-guessing — review conclusions are FACT from captured evidence, never "looks fine"), §11.4.40 (the full-suite retest is a later gate; §11.4.125 is the pre-build review gate that precedes it), §11.4.43 (TDD-fix — remediation lands RED-first), §11.4.50 (deterministic-consistency of the tests the review validates), §11.4.70 / §11.4.20 (the review IS a subagent dispatch — structurally-separated reviewer eliminates self-review blind spots), §11.4.92 (multi-pass change-evaluation is the author-side pass; §11.4.125 is the independent-reviewer pass), §11.4.102 (any defect the review finds triggers systematic-debugging before remediation), §11.4.107 (the review verifies AV/test-validation analyzers are not themselves bluffs), §11.4.108 (the review checks each fix's runtime signature is declared + verifiable), §11.4.110 (the build-readiness verdict + clash detector are mechanical gates the review complements with judgement).

Enforcement. Propagation gate `CM-COVENANT-114-125-PROPAGATION` enforces the literal anchor `11.4.125` across the consumer fleet (every CLAUDE.md / AGENTS.md / QWEN.md). Recommended gate `CM-CODE-REVIEW-GATE-BEFORE-BUILD` (the build orchestrator records a code-review-completed marker — with the review's findings + their remediation-or-no-blocking-findings determination — produced AFTER the batch's last fix and BEFORE the pre-build sweep + main build; a build that starts without a fresh review marker for the current batch FAILs).

Paired §1.1 meta-test mutations: strip the 11.4.125 literal → propagation gate FAILs; start a build with no fresh code-review marker for the batch → CM-CODE-REVIEW-GATE-BEFORE-BUILD FAILs. (Gate-code implementation lands as a separate follow-up work item; this anchor defines the contract.)

Classification: universal (§11.4.17) — "an independent multi-layer code-review-agent gate analyzing all work + data + codebase + git history for quality/safety/will-it-really-work + genuine test-validation, with findings blocking the build, runs BEFORE the pre-build sweep + main build" is a platform-neutral discipline reusable by ANY project that builds an artifact from source; the consuming project supplies its concrete review-marker mechanism + build-orchestrator integration per §11.4.35.

Canonical authority: this Constitution.md §11.4.125 in the HelixConstitution submodule. All consuming projects restate + cite via §11.4.35 inheritance.

Non-compliance is a release blocker regardless of context. No escape hatch — no `--skip-code-review`, `--build-without-review`, `--no-review-gate`, `--review-optional`, `--trust-the-author` flag exists.

2026-06-08

§ . . — Code-review iterate-until-GO
+ rock-solid-evidence mandate (User mandate,
- -)

Forensic anchor — verbatim user mandate (2026-06-08): "For any fixes/changes given back to us for re-work by the code-review process, once we fix/improve everything per the code-review's requests, we MUST RE-RUN code-review AGAIN until we get a GO from it with NO new issues reported or warnings of any kind! All results produced by this whole process MUST ALWAYS give us rock-solid PHYSICAL evidence that the fixed/improved codebase really works now as expected, with no false results and no bluff(s) of any kind."

When the §11.4.125 code-review returns ANY finding — BLOCKING, nit, or warning — and the author fixes/improves the batch per that review, the code review MUST BE RE-RUN, and MUST KEEP being re-run after each remediation round, until it returns a clean GO with ZERO new issues AND ZERO warnings of any kind. A single pass that "addressed the findings" is NOT sufficient: the corrected batch MUST pass a FRESH adversarial review (a re-review can surface NEW findings

introduced by the very fixes that closed the prior ones — the §11.4.1 fix-A-creates-B failure mode). The loop terminates ONLY on a clean GO (no new findings, no warnings); a residual warning is itself a finding that re-arms the loop.

Every round's verdict AND every fix's validation MUST carry rock-solid PHYSICAL captured evidence per §11.4.5 / §11.4.69 / §11.4.107 (captured audio / video / sysfs / dumphsys / sink-side / runtime-signature) proving the fixed/improved codebase REALLY works as expected on the target — never metadata-only / configuration-only / absence-of-error / grep-without-runtime. No false results, no bluff of any kind, at any round. A reported GO unbacked by captured physical evidence is itself a §11.4 PASS-bluff at the review-loop layer.

§11.4.134 REFINES / STRENGTHENS §11.4.125 (which requires the review to iterate "until no blocking findings remain"): it makes the loop EXPLICIT (re-run after every remediation round, not once), raises the termination bar to ZERO findings AND ZERO warnings (not merely zero-blocking), and BINDS rock-solid physical evidence to every round (not just the final one). Classification: universal (§11.4.17). Composes §11.4.125 / §11.4.1 / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.69 / §11.4.107 / §11.4.50 / §11.4.108 / §11.4.123. Propagation gate CM-COVENANT-114-134-PROPAGATION (literal 11.4.134) + recommended gate CM-CODE-REVIEW-ITERATE-UNTIL-GO + paired §1.1 meta-test mutation (gate-code = separate work item).

Canonical authority: this Constitution.md §11.4.134 in the HelixConstitution submodule. All consuming projects restate + cite via §11.4.35 inheritance.

Non-compliance is a release blocker regardless of context. No escape hatch — no `--skip-rereview`, `--single-review-pass`, `--warnings-ok`, `--evidence-optional` flag exists.

2026 06 09

§ . . — Universal code-review
mandate — every change reviewed, always, no
exception (User mandate, — —)

Forensic anchor — verbatim user mandate (2026-06-09):

"ALL changes we do MUST pass through the code review step!!! ALWAYS!!!"

EVERY change made to ANY repository this Constitution governs — without exception — MUST pass through an INDEPENDENT code-review step BEFORE it is accepted, committed, or built. There is NO change class exempt: source code, fixes, tests, gates, meta-test mutations, documentation, doc-tooling, build/CI scripts, configuration, governance files (Constitution / CLAUDE / AGENTS / QWEN), conductor main-stream edits, sub-agent output, refactors, single-line edits — all of it. This is the ABSOLUTE form of §11.4.125 (which mandated a code-review agent gate after a batch of fixes, before the pre-build sweep + main build): §11.4.142 removes every implicit scoping that §11.4.125's "after all fixes/changes/ implementations are done" phrasing could be read to leave open — there is no "this is just a doc edit" exemption, no "this is just a one-liner" exemption, no "the author already self-reviewed (§11.4.92)" substitution, no "trivial change" carve-out. If a diff exists, it gets an independent review.

Independence is load-bearing. The review MUST be performed by a reviewer structurally separated from the author (a dedicated code-review agent, subagent-driven by default per §11.4.70 / §11.4.20, or a distinct human reviewer) — NOT the author re-reading their own work. §11.4.92's multi-pass self-evaluation is the AUTHOR-side discipline and remains required, but it does NOT satisfy §11.4.142: the whole point of an independent review seam is to catch the blind spots self-review cannot (the structurally-separated reviewer seam §11.4.70 names). A self-review labelled as the code-review step is a §11.4.142 violation of the same class as a §11.4 PASS-bluff at the review layer.

The review is itself anti-bluff. A rubber-stamp "looks good" with no examined findings is not a review (§11.4 / §11.4.1) — the review MUST genuinely analyse the change for correctness, safety (no host §12 / data §9 / target-hardware §11.4.133 regression), will-it-really-work (no solve-A-create-B per §11.4.1), end-user behaviour (§11.4 / §107), and test-genuineness (every covering test exercises the work and catches its negation per §1.1). Its conclusions are captured evidence per §11.4.5 / §11.4.69. **The review iterates to a clean GO per §11.4.134** — any finding (BLOCKING / nit / warning) re-arms the loop; the change is accepted only after a re-run review returns ZERO new findings AND ZERO warnings, with rock-solid physical evidence per §11.4.134.

Composition, not duplication. §11.4.142 is the universal "always, every change, no exception" capstone over the code-review family: §11.4.125 supplies the multi-layer review CONTENT (the six analysis dimensions + the build-gate marker CM-

CODE-REVIEW-GATE-BEFORE-BUILD); §11.4.134 supplies the iterate-until-GO + rock-solid-evidence LOOP; §11.4.92 supplies the author-side self-review that PRECEDES (never replaces) the independent review; §11.4.70 / §11.4.20 supply the subagent-driven default execution model. §11.4.142 binds them and strips the last exemption: the review step is not a batch-end nicety, it is a precondition on the ACCEPTANCE of any diff. Honest boundary (§11.4.6): a passing review proves the change was independently examined and found sound to the reviewer's evidence — it does NOT replace the §11.4.108 four-layer runtime-signature verification nor the §11.4.40 full-suite retest; it is one of MULTIPLE STRONG LAYERS, and the FIRST one every change crosses.

Forensic FACT (teaching citation, 2026-07-22 — review-before-push is not optional). A transport-hardening change was committed AND pushed to a project's `main` BEFORE any independent review (the author honestly flagged the omission); the post-hoc §11.4.209-substrate review then returned CONDITIONAL/NO-GO with ONE BLOCKING finding — the change had left a security-relevant protection default in the unsafe direction (an operator-keystroke-obscuring option shipped default-off with no disclosure) — plus two Important and four Minor findings, and the remediation consumed a full fix-forward cycle (RED-first test updates per §11.4.115, the launcher fix, three documentation surfaces, and a §11.4.134 re-review) that a pre-push review would have folded into the original change. This is the exact class this anchor forbids: the review found a real security-default defect that was ALREADY public on every mirror when it was found. The review comes BEFORE commit/push; a post-hoc review that catches a Blocking issue PROVES the mandate, it is not a substitute for it (remediation is fix-forward per §11.4.113 — never a history rewrite).

Classification: universal (§11.4.17) — a platform-neutral acceptance-gate discipline reusable by ANY project; the consuming project supplies its concrete reviewer-dispatch mechanism + the change-acceptance seam (commit wrapper / merge queue / PR gate) per §11.4.35. Composes §11.4.1 / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.20 / §11.4.40 / §11.4.69 / §11.4.70 / §11.4.92 / §11.4.108 / §11.4.110 / §11.4.125 / §11.4.134 / §107 / §1.1. Propagation gate `CM-COVENANT-114-142-PROPAGATION` (literal `11.4.142` across the consumer fleet) + recommended gate `CM-EVERY-CHANGE-REVIEWED` (every accepted change carries a fresh independent-review-completed marker for its diff, produced by a reviewer distinct

from the author, before acceptance/commit/build) + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; accept a diff with no independent-review marker → CM-EVERY-CHANGE-REVIEWED FAILs; gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.142. Non-compliance is a release blocker regardless of context. No escape hatch — no --skip-review, --no-review, --trivial-change-exempt, --doc-edit-exempt, --self-review-suffices, --review-after-commit flag exists.

§ . . — Independent multi-angle impact-research per change (User mandate, — —)

Forensic anchor — verbatim operator intent (2026-06-10): for EVERY fix / change / new feature, independent agents MUST research potential issues related to that work AND connected/dependent features, from VARIOUS ANGLES, to be SURE no new issues are created, no existing working functionality is broken, and no dangerous code is introduced "like we had with the recents feature", incorporating a VARIETY OF TOOLS.

Forensic anchor (FACT, project-internal): the recents / 3-button-nav path shipped a latent AIDL-interface-contract mismatch that PASSED code review (the diff "looked correct") yet failed at RUNTIME ONLY — it ANR'd on a recents-reaping gesture — because no review angle interrogated the interface contract, the concurrency/lifecycle, or the connected feature it silently broke; the §11.4.108 SOURCE → ARTIFACT → RUNTIME → USER-VISIBLE gap, here caught only after the fact.

§11.4.145 shifts blast-radius discovery LEFT: for EVERY change, INDEPENDENT impact-research agents (subagent-driven §11.4.70/§11.4.20, structurally separate from the author — §11.4.92 self-eval PRECEDES, never satisfies — adversarial "refute-the-change" stance to defeat the documented LLM confirmation-bias failure mode where reviewers accept plausible-but-wrong arguments) research the change AND its connected/dependent features across a CLOSED SET OF EIGHT ANGLES — (1) **correctness/logic**, (2) **regression** (what existing working feature could break — via call-graph impact enumerating every direct + transitive caller and contract-dependent feature, each mapped to its test), (3) **latent/dangerous-code**

— **the recents class** (runtime-only failure, interface/ABI/contract mismatch, concurrency/data-race, resource lifecycle), (4) **security** (taint/injection/secret/unsafe-API), (5) **performance** (latency/memory/thermal under load vs baseline), (6) **host/data/target-hardware safety** per §12/§9/§11.4.133, (7) **cross-feature interaction** (shared state/timing/hardware contention), (8) **business-logic conformance** (matches the spec AND the connected features' contracts, not merely compiles).

Each angle MUST name the TOOL(S) used + obtain the required DEPTH of codebase + business-logic detail (read the diff, the full changed unit, its declared contract, the spec section, and every connected feature — NEVER the diff lines alone) + emit a captured-evidence conclusion per §11.4.5/§11.4.69 (the tool's findings/impact/diff artefact path — "looks fine" is forbidden, §11.4.1). A genuinely-N/A angle is recorded `NOT_APPLICABLE: <reason>` per §11.4.6, never silently skipped. The output is a per-change impact-research REPORT (one section per angle: tool + evidence path + verdict) + a single GO/NO-GO that BLOCKS acceptance/commit/build on ANY unmitigated risk in ANY angle; the change is fixed/mitigated and the affected angles re-researched, iterating to a CLEAN GO (zero unmitigated risk + zero warning, rock-solid physical evidence) per §11.4.134 before the §11.4.125 review gate.

Honest boundary (§11.4.6): a GO proves cross-angle internal-consistency + bounded blast-radius — it does NOT replace §11.4.108 four-layer runtime-signature verification nor the §11.4.40 full-suite retest; it is one of MULTIPLE STRONG LAYERS, the FIRST research pass every change crosses.

Classification: universal (§11.4.17) — the consuming project supplies its concrete toolkit (static-analyzer, call-graph engine, interface-freeze mechanism, behavioral harness) + per-angle depth per §11.4.35. Composes/strengthens §11.4.92 (adds independent tool-backed angle research) / §11.4.125 (its impact-research input) / §11.4.108 (shifts blast-radius discovery left) / §11.4.110 (angles 2/3/7 use the build-graph + interface-freeze) / §11.4.134 (iterate-to-GO) / §11.4.78 / §11.4.79 (call-graph impact engine) / §11.4.107 / §11.4.85 (behavioral angle) / §11.4.133 / §12 / §9 (angle 6) / §11.4.99 (cite latest sources) / §11.4.6 / §1.1. Propagation gate `CM-COVENANT-114-145-PROPAGATION` (literal `11.4.145` across the consumer fleet) + recommended gate `CM-IMPACT-RESEARCH-PER-CHANGE` (every accepted change carries a fresh independent impact-research report covering all eight angles with a clean GO, produced by agents distinct from the author, before acceptance/commit/

build) + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; accept a change with a report missing an angle or with an unmitigated NO-GO → CM-IMPACT-RESEARCH-PER-CHANGE FAILs; gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.145. Non-compliance is a release blocker. No escape hatch — no

`--skip-impact-research` , `--single-angle-suffices` , `--author-researches-own-change` , `--diff-skim-OK` , `--no-go-overridable` , `--trivial-change-no-research` flag.

2026 06 21

§ . . — Universal Independent Verification Agent Mandate (User mandate, — —)

Forensic anchor — direct user mandate (2026-06-21).

The mandate. Every code change OR recorded media artifact produced by any project governed by this Constitution MUST pass through an INDEPENDENT verification agent structurally separate from the author (§11.4.70/§11.4.20, dedicated subagent or distinct human, NEVER the author re-reading their own work — §11.4.92 self-evaluation PRECEDES and never satisfies). The verifier is one of MULTIPLE STRONG LAYERS — the FIRST verification layer every output crosses, complementing (never replacing) §11.4.108 runtime-signature verification, §11.4.125/§11.4.142 code review, and §11.4.40 full-suite retest.

(a) For CODE changes. The independent verification agent MUST: (i) run the §11.4.142 code-review discipline (adversarial analysis, blast radius, contract integrity, test genuineness), (ii) build and test the changed packages (invoke the project's `pre_build_verification.sh` on the diff scope), (iii) run the paired §1.1 meta-test mutations for every gate the change touches (asserting each mutated gate still FAILs and its restoration still PASSes), (iv) verify the §11.4.108 runtime-signature registry entry for this change declares one machine-checkable observable. A code change that passes only one of (i)–(iv) is NOT verified — all four must hold.

(b) For MEDIA artifacts (video/audio/screenshots/asciicast/text output). The independent verification agent MUST: (i) run the §11.4.163 media validation pipeline on every recorded artifact, (ii) confirm the content is genuine — real LLM/

service responses, not mocked/stub/simulated/placeholder (per §11.4.159(I)), (iii) confirm the artifact format is correct (MP4 H.264 for video, WAV/FLAC for audio, PNG for screenshots, cast+mp4 for asciicast per §11.4.159(B)), (iv) cross-check the evidence path against the artifact (the path exists, is non-empty, and the content matches the feature's expected output). A media artifact that passes format but shows simulated/placeholder content is a finding, not a PASS.

(c) For DOCS changes. The independent verification agent MUST: (i) validate that HTML + PDF + DOCX exports exist and are current per §11.4.65 (mtime parity), (ii) verify the §11.4.44 revision header is present and its integer has incremented from the previous commit, (iii) for docs that carry a §11.4.86 drift-proof fingerprint, verify the fingerprint matches the sorted roster/keyset. A doc change with a stale export or missing revision header is NOT verified.

(d) For CONFIG changes. The independent verification agent MUST: (i) validate YAML/JSON/TOML syntax (parse with the language's own parser — `yq` , `jq` , basic JSON parser, `python -m json.tool` , etc.), (ii) cross-reference against a schema where available (e.g. JSON Schema for `.settings.json` , YAML Schema for `helix-deps.yaml`), (iii) confirm no credential or secret pattern leaked per §11.4.10. A config change with a syntax error, schema violation, or leak is a finding, not a PASS.

(e) Structured findings. The verifier MUST produce a structured findings report — for each finding: item (which code/media/doc/config), verification class (a–d), tool/command used, evidence path (log file, build output, mutation result, validator verdict), PASS/FAIL. "Looks good" without evidence is not a finding — a rubber-stamp verifier is a §11.4 PASS-bluff at the verification layer.

(f) Iterate to GO. If any finding exists (PASS/FAIL indeterminate, BLOCKING, or even a warning), the verifier iterates per §11.4.134 — the author fixes the finding, the verifier RE-VERIFIES, until GO with ZERO findings and ZERO warnings. A single-pass verifier that reports findings but accepts GO because "the findings were minor" is a §11.4.165 violation.

(g) Anti-bluff — verifier self-validation. The independent verification agent is itself anti-bluff. The verifier's meta-test is a paired §1.1 mutation that introduces a deliberately broken code change (e.g. a gate with a stripped assertion) and asserts the verifier catches it. A verifier that passes a known-broken change is a bluff gate and must be investigated per §11.4.102 before any further use.

Honest boundary (§11.4.6). The independent verification agent confirms code/media/docs/config integrity at the source/artifact layer — it does NOT prove the feature works for the end user on the deployed target (that rests on §11.4.108 runtime-signature + §11.4.130 validate-fix-first + §11.4.40 full-suite retest) and does NOT replace the §11.4.40 full-suite retest or §11.4.118 discovery pass. A verifier that reports GO on an artifact whose downstream deployment is broken is not a verifier failure — that is caught by the later §11.4.108 layer. A verifier that reports GO without having actually run any tool (rubber-stamp) IS a verifier failure and a §11.4.165 violation.

Classification: universal (§11.4.17) — the consuming project supplies its concrete build+test command per diff scope, mutation-test runner, media validator invocation, doc export checker, config parser, and verifier-dispatch agent per §11.4.35. Composes

§11.4.142/.163/.44/.65/.86/.108/.125/.134/.10/.40/.92/.70/.20/.102/.159(I)/.159(B)/
 §1.1. Propagation gate CM-COVENANT-114-165-PROPAGATION (literal 11.4.165)
 + recommended gate CM-INDEPENDENT-VERIFICATION-AGENT (every output — code/media/docs/config — passes an independent verifier structurally separate from the author, produces structured findings with evidence paths, iterates to zero-finding GO; verifier is self-validated by a paired §1.1 mutation) + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; accept an output without the independent verifier's GO → CM-INDEPENDENT-VERIFICATION-AGENT FAILs; verifier reports GO on a known-broken fixture → self-validation FAILs; gate-code = separate work item).

2026 07 14

§ . . — Exhaustive all-scenario,
 all-angle code-review + verify-against-captured-
 runtime-evidence mandate (User mandate,
 — —)

Forensic anchor — verbatim user mandate (2026-07-14):

"CRITICAL: Code review MUST TAKE into the accounts all scenarios, look at the problem or change from all angles, deepest possible analysis! No gaps, shortcomings or false results or bluff is allowed! Extend our root constitution Submodule and all relevant files properly so the review process is more precise and we do not have situations like the last one!"

Forensic case study (genericised, 2026-07-14 — "the last one"). A code-review (§11.4.125) returned a clean **GO with ZERO findings** on a boot-crash fix. It dismissed a defensive clamp branch as an INFORMATIONAL NIT — "effectively unreachable ... the input is always valid" — WITHOUT PROVING that assumption against the actual runtime input space. The fix shipped, was deployed to the target, and the target STILL crash-looped: the degenerate input the review had assumed could not happen DID happen at runtime (an empty / default-initialised profile opened with the exact zero / default value the review assumed impossible). The crash was a MULTIPLICATIVE size computation — a product $A \times B$ (e.g. `frame_size = channels × bytes_per_sample(format)`) — with TWO INDEPENDENT zero-triggers; the review verified only ONE factor (`A`, e.g. `channels`) and ASSUMED THE OTHER (`B`, e.g. `format`) away. The crash-loop reproduced on the deployed target under conditions PRESENT in the pre-fix crash evidence that the review never cross-checked the fix against — the evidence was available and not consulted.

Every code-review this Constitution governs (STRENGTHENING §11.4.125 / §11.4.134 / §11.4.142 — the review remains NECESSARY; §11.4.194 makes it EXHAUSTIVE-and-evidence-grounded so a clean GO can no longer be returned on an unproven assumption) MUST, before returning GO:

(1) Enumerate the FULL input / scenario space of BOTH the change AND the defect it addresses — every dimension, every code path, every interacting feature, every input domain the change touches — NOT merely the reported / happy-path scenario. For a MULTI-FACTOR failure mode (e.g. a product of `N` terms, ANY of which being degenerate — zero, empty, null, default-initialised, out-of-range — triggers the failure), the review MUST enumerate AND INDEPENDENTLY VERIFY the fix covers EACH factor / dimension. Verifying ONE factor and ASSUMING the rest are safe is the EXACT forbidden gap this anchor closes (the forensic `channels` -verified / `format` -assumed pattern).

(2) Prove every assumption. "This can't happen" / "unreachable" / "always valid" / "the caller never sends X" / "the input is always in range" MUST be PROVEN from the code + spec + captured evidence — NEVER accepted on faith, NEVER inferred from "it usually is" (§11.4.6 no-guessing). An unproven assumption that later proves false is a §11.4 REVIEW-LAYER bluff of the SAME severity class as a PASS-bluff. A review NIT that DISMISSES a path / branch as unreachable / dead / defensive-only MUST carry the PROOF of unreachability (the callers enumerated, the input domain bounded, the degenerate value shown genuinely impossible) — a bare "effectively unreachable" NIT with no proof is ITSELF a finding, not a dismissal.

(3) Verify the fix against ANY captured runtime evidence. When captured runtime evidence of the failure EXISTS — a crash log / tombstone / stack trace / core dump / sink-probe report / the §11.4.108 runtime-signature / a reproducer — the review MUST confirm the fix ACTUALLY PREVENTS the CAPTURED failure (walk the fix against the EXACT captured conditions, not a hypothetical). A review that GO's a fix which then FAILS at runtime on a scenario PRESENT in the evidence (or a scenario the review CHOSE NOT TO CONSIDER) is a bluff — the evidence was available and not cross-checked.

(4) All angles. The review MUST cover ALL of: correctness against the stated goal; the FULL input domain (every factor of a multi-factor computation, every boundary, every degenerate value); every downstream / interacting feature — ESPECIALLY must-not-break constraints (per §11.4.133 target / hardware safety, and any feature the change shares state / timing / a resource with — e.g. audio / multichannel / passthrough integrity); the TESTS' OWN coverage of every dimension (a test that checks only ONE factor of a multi-factor bug is ITSELF a coverage gap the review MUST catch and require closed); and the §1.1 paired mutation covering EACH dimension (a mutation that flips only one factor leaves the others un-guarded).

(5) Honest boundary (§11.4.6). "Exhaustive" means the ENUMERATED + EVIDENCE-GROUNDED scenario space — NOT a claim of literal omniscience. A dimension that genuinely cannot be analysed (external state the review cannot inspect, a runtime path with no reproducer) MUST be stated as an EXPLICIT gap (UNCONFIRMED: / UNKNOWN: / PENDING_FORENSICS: with a tracked follow-up) — NEVER silently assumed safe. A silently-assumed-safe un-analysed dimension is the §11.4.194 violation.

(6) The quality-harvest review classes (research-derived extension, 2026-07-23 — operator mandate: every check, diagnostic method, and preventive measure from the reopen/first-touch root-cause forensics is checked through EVERY code-review process). In addition to (1)–(5), every review MUST explicitly CHECK, per reviewed change: **(a) closure-evidence class vs defect layer (§11.4.226)** — no terminal or ready-class status in the change's scope moves on evidence whose class ranks below the defect layer's floor; a user-visible property "verified" by file-existence checks plus greps, an echo-assertion, or a state-only repair closing a defect as fixed is a FINDING; **(b) needed measurements** — every load-bearing measurement the change or its tests performs satisfies §11.4.201(6)–(12) (a reported absence cites its class-matched control needle), and NO count was converted into a finding without reading the underlying lines — a COUNT IS A LEAD, THE LINES ARE THE FINDING (forensic FACT: twice in one measured cycle a count was read as state — a "mixed half-written doc" that was actually coherent with a labelled historical citation, and an inflated FAIL total that included the summary line counting itself); **(c) governance-and-gate integrity (§11.4.227)** — a change that NAMES a gate lands its implementation or a registered deferral in the same change, and a governance edit preserves anchor-block integrity (exactly-once per mirror, lockstep-identical, no anchor-number collision); **(d) reviewer-authored adversarial mutations** — the author's self-authored mutation set is NECESSARY but NOT SUFFICIENT validation of a guard/test: forensic FACT (from the consuming project's dated forensic corpus cited in the 2026-07-22 root-cause analysis): an author's six self-authored mutations were ALL caught while an independent reviewer's FIRST THREE different mutations ALL sailed through, each restoring a user-visible defect. The reviewer MUST attempt at least one mutation the author did not write (or record honest infeasibility per §11.4.3); a guard whose reviewer-authored mutation survives is unvalidated instrumentation per §11.4.115(F) and a FINDING.

Classification: universal (§11.4.17) — a platform-neutral review-rigor discipline reusable by ANY project; the consuming project supplies its concrete input registries, must-not-break constraint set, and captured-runtime-evidence sources per §11.4.35. Composes §11.4.125 (the code-review-agent gate this strengthens) / §11.4.134 (iterate-until-GO — a §11.4.194 gap is a finding that re-arms the loop) / §11.4.142 (every-change-reviewed — §11.4.194 raises the review's PRECISION bar) / §11.4.6 (no-guessing — the operative discipline behind clauses 2 + 5) / §11.4.108 (runtime-signature — the captured evidence clause 3 cross-checks

against) / §11.4.130 (validate-the-fix-first-after-redeploy — the runtime confirmation that would have caught "the last one") / §11.4.107(10) (self-validated analyzer — the multi-factor coverage discipline applied to the review's OWN test-coverage check). Propagation gate `CM-COVENANT-114-194-PROPAGATION` (literal `11.4.194` across the consumer fleet) + recommended gate `CM-EXHAUSTIVE-REVIEW-ALL-SCENARIOS` (a review GO must record the enumerated scenario / factor space + the proof of every dismissed-as-unreachable branch + the cross-check against captured runtime evidence where it exists) + paired §1.1 meta-test mutation (strip the enumeration / the assumption-proof / the captured-evidence cross-check from a review record → the gate FAILs; strip the literal → the propagation gate FAILs; gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.194.

Non-compliance is a release blocker regardless of context. No escape hatch — no

```
--partial-scenario-review , --assume-unreachable-OK ,
--skip-captured-evidence-crosscheck ,
--single-factor-review-suffices , --dismiss-branch-without-proof flag.
```

2026 09 15

§26 07 15 . . — Code-review MUST run on the Opus model at xhigh effort (Fable escalation for big/complex reviews) (User mandate, — — , supersedes — —)

Verbatim operator mandate (2026-09-15), superseding the 2026-07-15

mandate below: the mandatory independent code-review MUST by default be performed with the **Opus model at xhigh effort**; for a review the operator or the reviewer itself classifies as **extremely big or complex**, the **Fable model at xhigh effort** MUST be used instead.

Prior mandate (2026-07-15), WITHDRAWN as current, kept for lineage: *"the mandatory independent code-review MUST ALWAYS be performed with the Fable model at xhigh effort; if Fable is unavailable, use Opus at xhigh effort."* That ordering is inverted by the 2026-09-15 mandate above: Opus-xhigh is now the DEFAULT reviewer substrate and Fable-xhigh the ESCALATION tier for large/complex work, not an unavailability fallback.

Every mandatory independent code-review the constitution requires — the §11.4.125 code-review-agent gate before pre-build + main build, the §11.4.142 universal every-change-review (no exception), and the §11.4.194 exhaustive all-scenario/all-angle review + verify-against-captured-runtime-evidence, iterated to a zero-finding GO per §11.4.134 — MUST be dispatched on the **Opus model at xhigh effort** by default. **(A) Escalation (closed, ordered)**. When the review is classified — by the operator, or by the dispatching agent using a stated, evidenced reason (§11.4.6, never a guessed classification) — as **extremely big or complex** (very large diff, cross-cutting/architectural blast radius, a governance/gate-authoring change, or any change the operator explicitly flags), the review MUST run on the **Fable model at xhigh effort** instead of Opus. xhigh effort is REQUIRED in both the default and the escalated case — the exhaustive §11.4.194 all-angle depth is not reachable at a lower effort. **(B) Independence preserved (§11.4.70/§11.4.20)**. The Opus-xhigh (or, on escalation, Fable-xhigh) model runs as the structurally-separated reviewer (never the author self-reviewing); §11.4.209 sets the MODEL + EFFORT of that review, it does not weaken the independence, the iterate-until-GO loop (§11.4.134), or the rock-solid captured-evidence requirement (§11.4.5/§11.4.69/§11.4.107). **(C) Honest boundary (§11.4.6)**. A review claimed "done" that actually ran on a weaker model / lower effort than the tier this anchor requires (Opus-xhigh by default, Fable-xhigh once classified big/complex) is a §11.4 bluff at the review-substrate layer — the review's captured evidence MUST record which model + effort actually performed it, and, when escalated, the stated reason the review was classified big/complex. **(D) Values already burned in a public repository (§11.4.10)**. Any plaintext credential value that has appeared in documentation, a spec, or commit history — including as a worked example — MUST be treated as compromised from the moment of commit; a document illustrating this anchor or any credential-adjacent mandate MUST NOT embed a real-looking username/password pair, even as an example, without the accompanying statement that the pair is compromised-by-disclosure and MUST NOT be relied on as a real secret. STRENGTHENS §11.4.125/§11.4.142/§11.4.194/§11.4.134 — it does NOT replace any of them; it pins the review's execution substrate.

Classification: universal (§11.4.17). Composes §11.4.125 / §11.4.134 / §11.4.142 / §11.4.194 / §11.4.6 / §11.4.10 / §11.4.20 / §11.4.70 / §11.4.196 (alias/model selection) / §11.4.231 (nano-precision model-tier selection — §11.4.209 remains the hard pin for review/merge-conflict scope that §11.4.231(E) explicitly does not

reach). Propagation gate `CM-COVENANT-114-209-PROPAGATION` (literal `11.4.209` across the consumer fleet) + recommended gate `CM-CODE-REVIEW-OPUS-XHIGH-DEFAULT-FABLE-ESCALATION` (the mandatory code-review's captured evidence records `model=Opus + effort=xhigh` by default, OR `model=Fable + effort=xhigh` with a recorded big/complex classification reason) + paired §1.1 meta-test mutation (record a code-review run on a lower-effort or non-Opus/non-Fable model without a recorded escalation reason → the gate FAILs; strip the literal → the propagation gate FAILs; gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.209.

Non-compliance is a release blocker. No escape hatch — no

```
1 1 4 2 4 0
--review-any-model , --review-lower-effort-OK , --skip-opus , --non-
xhigh-review , --escalate-without-recorded-reason flag.
```

§ 11.4.209 — **Producer ≠ Verifier scope separation:** the actor that produces a change/artifact/verdict cannot be the one that gates it — **least privilege over cleverness** (research-derived, — —)

Compact summary: every project MUST enforce STRUCTURAL SEPARATION between the actor that PRODUCES a change/artifact/verdict and the actor that GATES that same change/artifact/verdict — the producing actor cannot write, weaken, delete, or edit the acceptance tests, the oracle, the gate configuration, the verdict store, the paired §1.1 mutation, or any other check whose green result is a precondition for the producer's own work being certified done. Enforcement is by CAPABILITY (the producer STRUCTURALLY cannot reach the gate) rather than by INSTRUCTION (the producer is TOLD not to touch the gate), and the enforcement mechanism is LEAST PRIVILEGE OVER CLEVERNESS (extraction file line 1488 — **MODULE ATTRIBUTION CORRECTED 2026-08-20: this citation originally read "module 27". The quoted passage has ZERO occurrences in**

`module_27.txt` and resolves in the tracked corpus to module 28 ("Anti-Bluff Verification & Release Validation"), verified locator `module_28:393` .

Instrument control-needed per §11.4.201(7)(b) before the absence was reported. — module 28, verified locator `module_28:393` : "the durable fix is least privilege over cleverness — the same logic as prompt-injection defense ... the load-bearing move is that the agent chasing the green gate cannot reach the gate, the

oracle, or the evidence store"). The failure mode this anchor forbids: a producer that CAN edit the tests/oracle/gate config/verdict store DOES edit them under long-horizon optimization pressure — hardcoding expected values, weakening assertions, mocking the unit under test, deleting the failing test, or writing `verified: true` on itself (extraction lines 5695, 5711-5717, 5722-5730, 1343, 1396, 1482-1488). §11.4.240 makes the separation a STRUCTURAL PROPERTY of the codebase and its automation, not a social one.

(A) THE FOUR SCOPES + THE SEPARATION MATRIX. Every change/artifact/verdict flows through four SCOPES that MUST be distinct actors under §11.4.240: (i) PRODUCER — the actor that writes the source diff, the config change, the migration, the artifact-generation code; (ii) ORACLE — the actor / mechanism that decides "correct output" for a given input (test author, specification writer, characterization-baseline capturer, reference-implementation author per §11.4.245 oracle-independence); (iii) GATE — the actor / mechanism that decides whether the change PASSES the check (CI configuration, gate scripts, custody-chain enforcer per §11.4.146(D3), release-verdict-coverage machinery per §11.4.135); (iv) VERIFIER — the actor / mechanism that RE-RUNS the gate from a clean checkout and attaches receipts (independent verifier per §11.4.165, §11.4.209 Fable-xhigh substrate for code review, §11.4.194 exhaustive review). The mandate: the PRODUCER of a change cannot ALSO be the ORACLE, GATE, or VERIFIER for that same change. Extraction line 5695 (module 34 core concept 4 — module attribution VERIFIED CORRECT 2026-08-20, verified locator `module_34:56` ; offset un-resolvable per the §11.4.239 CITATION NOTE): "the agent that wrote the code cannot be the authority that certifies it — self-preference bias and shared context make self-review worthless. Verification comes from an independent instance (fresh context, read-only, re-running from a clean checkout) and from an oracle the producer did not author and cannot edit ... Remove any one property and it degrades to self-certification." A four-scope collapse (one actor plays all four roles) is the maximum-severity form; any two-scope collapse involving the producer is a §11.4.240 finding of §11.4/§11.4.1 severity at the certification layer.

(B) LEAST PRIVILEGE OVER CLEVERNESS — CAPABILITY, NOT INSTRUCTION. Separation MUST be enforced by structural CAPABILITY LIMITS, not by prompts / instructions / policy documents / social contracts asking the producer "please do not touch the tests." Extraction lines 1482-1488 [MODULE CORRECTED 2026-08-20 — the (E) Extraction-sources line attributes these offsets to module 27; the quoted passage resolves to module 28, verified locator

module_28:383] name the load-bearing mechanism: "The producer has no write access to the test files, the oracle, the CI/gate config, or the verdict store — enforced by scope, not by asking. If it can't touch the gate, it can't game it." Concrete enforcement channels the consuming project MUST supply per §11.4.35: (1) FILESYSTEM SCOPE — the producer's write scope excludes tests/ , oracle/ , .github/workflows/ , scripts/gates/ , verdicts/ (or their consumer-declared equivalents); (2) BRANCH / REVIEW PROTECTION — protected paths (see §11.4.248 PROTECTED-SPEC + CODEOWNERS from the sibling 2026-08-15 operator decision) require an actor OTHER than the change author to approve modifications; (3) VERDICT-STORE ISOLATION — the store that records RED/GREEN verdicts per §11.4.115(F) is written only by the harness that ran the test, never by the producer's own code path; (4) EXECUTION ISOLATION — the verifier runs in a CLEAN CHECKOUT with FRESH CONTEXT (extraction lines 3168, 3117, 2381 [MODULE CORRECTED 2026-08-20 — the (E) Extraction-sources line attributes these offsets to module 34; the quoted passage has ZERO occurrences in module_34.txt and resolves to module 29 / module 30, verified locators module_29:655 , module_30:591 , module_30:655]: "an independent verifier re-running in a clean checkout") — the verifier does NOT inherit the producer's dirty worktree, cached artifacts, or in-flight session state. "The agent will not do that" is NEVER acceptable enforcement — extraction line 5715 [module 34 attribution VERIFIED CORRECT 2026-08-20, verified locator module_34:87] catalogues the failure ("We told it in the prompt not to hardcode / not to skip tests" as the shallow / red-flag answer). Exhortation is not capability; §11.4.240 refuses exhortation as the enforcement layer.

(C) THE THREE FORBIDDEN PRODUCER ↔ SCOPE COLLAPSES. The following collapses are §11.4.240 violations regardless of context, scale, or claimed rigor: (1) PRODUCER=ORACLE — the same actor writes the source diff AND the test that asserts its correctness, with no independent oracle authoring the assertion; the resulting test asserts whatever the code happens to do ("tautological testing trap," extraction line 5764 [module 34 attribution VERIFIED CORRECT 2026-08-20, verified locator module_34:155], referenced) rather than what the specification requires. Countermeasure: the oracle is authored FROM the specification / from a characterization baseline captured BEFORE the change (§11.4.243) / from a reference implementation (§11.4.245 derived oracle), not co-authored with the diff. (2) PRODUCER=GATE — the same actor writes the source diff AND has write access to the gate configuration / CI pipeline / the mutation-testing thresholds / the

paired §1.1 mutations; the producer can silence failing gates or weaken thresholds. Countermeasure: the gate configuration lives in a §11.4.248-protected path, and modifications require a non-producer approver. (3) PRODUCER=VERIFIER (self-certification) — the same actor produces the change AND writes the "verified: true" / "PASS" verdict for the change; the verdict is a producer-set flag with no independent re-run. Countermeasure: verdicts are MACHINE-WRITTEN by the harness that ran the test per §11.4.115(F), keyed to the target's artifact fingerprint read at run time, and the release-verdict-coverage seam per §11.4.135 REFUSES a release tag while any producer-asserted (rather than harness-written) verdict is in the release scope. Extraction lines 1212, 1343, 1396, 1631 [MODULE CORRECTED 2026-08-20 — attributed to module 27 in the (E) Extraction-sources line; the quoted passage resolves to module 28, verified locator `module_28:278`]: "Tamper-evident and outside the producer's reach — the producer of the change cannot edit the verdict or the oracle to make itself green."

(D) COMPOSITION WITH §11.4.115(F) / §11.4.125 / §11.4.135 / §11.4.142 / §11.4.146(D3) / §11.4.165 / §11.4.194 / §11.4.209. §11.4.115(F) machine-written verdict pairs are the (C)(3) countermeasure at the verdict layer — a producer-typed status label is not a verdict. §11.4.125's code-review-agent gate before pre-build + main build IS the (A) VERIFIER role at the review layer, and §11.4.142's universal every-change-review mandate + §11.4.209's Fable-xhigh review-substrate pin (Opus-xhigh fallback) provide the independence + substrate requirements. §11.4.135's standing regression-guard suite is the GATE layer at the release seam. §11.4.146(D3) status custody is the STATUS-WRITE-SEAM enforcement — the store REFUSES a producer-asserted terminal-status write without the full custody chain. §11.4.165 (universal independent verification agent) is the VERIFIER role generalised across every output. §11.4.194 (exhaustive all-scenario review) is the depth requirement applied to the verifier's pass. §11.4.240 does NOT invent a new mechanism at any of these layers — it ARTICULATES the property (structural producer-vs-verifier separation) that the existing mechanisms collectively enforce, and it forbids the collapses (§11.4.240(C)) that would render those mechanisms decoration.

(E) HONEST BOUNDARY (§11.4.6). Structural separation forbids the CLASS of self-certification failure modes; it does NOT prove the verifier is correct (a mis-calibrated verifier stays §11.4.107(10) golden-good/golden-bad territory; extraction line 1512 [MODULE CORRECTED 2026-08-20 — implicitly module 27 via the (E) Extraction-sources line; the quoted passage resolves to module 28, verified locator

`module_28:401`]: "a flaky or false-positive gate destroys the whole evidence chain above it"), does NOT prove the oracle is correct (§11.4.245 mandates oracle-strategy naming + independence), and does NOT guarantee the producer never tries to game the gate (it forbids the producer from succeeding at gaming the gate through direct write access; other gaming vectors — hardcoding to visible inputs, over-fitting the seen cases — are countered by §11.4.245 held-out oracles + property/metamorphic/differential coverage). Composes §11.4.6 (no guessing — the producer's claim is a prediction, only the verifier's captured re-run is evidence) / §11.4.20 (subagent-driven — the verifier IS the natural subagent dispatch) / §11.4.70 (subagent-driven default) / §11.4.92 (multi-pass evaluation — self-review PRECEDES independent review, never satisfies it) / §11.4.107(10) (self-validated analyzers — the oracle carries its own golden fixtures) / §11.4.115(F) (machine-written verdicts) / §11.4.125 (code-review-agent gate) / §11.4.134 (iterate-to-GO — the verifier's clean-GO terminates the loop) / §11.4.135 (standing regression-guard suite) / §11.4.142 (universal every-change-review) / §11.4.146(D3) (status custody chain refuses producer-asserted terminals) / §11.4.165 (universal independent verifier agent) / §11.4.194 (exhaustive review) / §11.4.201 (a gate that false-refuses a clean producer is a FAIL-bluff, so verifier honesty binds both directions) / §11.4.205 (seam-not-prose enforcement) / §11.4.209 (Fable-xhigh substrate pin for the review verifier) / §11.4.245 (oracle-strategy identification, independent by the same principle) / §11.4.248 (PROTECTED-SPEC + CODEOWNERS enforcement — the operator-decided mechanism for (B)) / §11.4.249 (four-role architecture + flight-recorder mechanism — §11.4.240 is the SEPARATION PRINCIPLE + capability enforcement, §11.4.249 is the concrete FOUR-ROLE {producer, oracle, gate, verifier} taxonomy + recorder that instantiates it; the two anchors mutually compose per §11.4.209 reviewer MINOR-1). Extraction sources [offsets un-resolvable per the §11.4.239 CITATION NOTE; module attributions AUDITED **and CORRECTED** 2026-08-20]: the offsets formerly attributed to "module 27" (1212, 1343, 1396, 1482-1488, 1498, 1613, 1631-1634) are a WRONG-MODULE attribution — every phrase this anchor quotes from them resolves to **module 28** ("Anti-Bluff Verification & Release Validation"), verified locators `module_28:278` (tamper-evident / the producer cannot edit the verdict or the oracle to make itself green), `module_28:383` (no write access to the test files; enforced by scope, not by asking; if it can't touch the gate, it can't game it), `module_28:393` (least privilege over cleverness; the same logic as prompt-injection defense), `module_28:401` (a flaky or false-positive gate destroys the whole evidence chain above it); **module 34** ("Quality Under AI-Assisted / Agentic Development") lines

5695, 5711-5717, 5722-5730, 3180, 3201, 2313 are module-CORRECT for every phrase quoted from them — verified locators `module_34:56` (the agent that wrote the code cannot be the authority that certifies it), `module_34:87` (we told it in the prompt not to hardcode), `module_34:155` (tautological testing trap); the offsets **3117-3168 and 2381**, listed under module 34, are a WRONG-MODULE attribution — the phrase quoted from them ("an independent verifier re-running in a clean checkout") has ZERO occurrences in `module_34.txt` and resolves to **module 29 / module 30**, verified locators `module_29:655` , `module_30:591` , `module_30:655` . **UNKNOWN (§11.4.6, stated rather than guessed):** offsets 1498, 1613, 1631-1634, 5711-5717, 5722-5730, 3180, 3201 and 2313 carry no quoted text in this anchor, so no phrase-level locator could be measured for them individually — their module attribution is UNVERIFIED: neither asserted correct nor asserted wrong.

(F) Independence-tier vocabulary + Tier-C capability-separation triggers, degrading honestly on a single-uid host (extension, spec-derived, speckit 002-anti-slop-enforcement, 2026-08-26). Clauses (A)–(E) state the SEPARATION PRINCIPLE and its capability-not-instruction enforcement in the abstract; this clause supplies the CLOSED VOCABULARY for recording HOW MUCH separation was actually achieved, and the two triggers that raise the required strength — because a project claiming "capability separation" where only "instance separation" is genuinely achievable is itself a §11.4.240 violation of the theatre this anchor exists to prevent. **(1) The closed tier set.** Per spec.md:124 (FR-035): every accepted verdict's recorded basis of independence MUST name which tier applied, drawn from the CLOSED SET {`instance`, `model`, `capability`} — `instance` (a separate agent/process instance verifies, sharing no in-flight state with the producer), `model` (a DIFFERENT model verifies — permitted, never required per (2) below), `capability` (a genuine host-level access-control boundary separates the producer's write scope from the verifier's — the strongest tier). **(2) Model diversity is permitted, never required — the false-positive guard.** Per spec.md:121 (FR-032): a verdict from an independent INSTANCE running the SAME model as the producer MUST be ACCEPTED on that basis alone (Tier B, permitted-never-required) — a check that refuses same-model independent verdicts has turned a legitimate separation into an unnecessary same-model prohibition, which is itself the exact class of false-positive refusal §11.4.201 forbids. This clause does NOT weaken the constitution's OTHER, distinct, narrower model-substrate pins — §11.4.256's requirement that a translation review run on a

DIFFERENT model/provider than the translator, and §11.4.209's requirement that every mandatory independent code-review run on the Fable model at `xhigh` effort (Opus `xhigh` on genuine Fable-unavailability); both remain in force unmodified for their own narrower domains (translation review; code review). FR-032's permitted-not-required framing governs only the GENERAL producer≠verifier case this anchor addresses, never those two domain-specific, stricter pins. **(3) Tier-C trigger 1 — the producer holds write access to the evidence/verdict store.** Per spec.md:122 (FR-033): at a seam where the producer holds write access to the evidence or verdict store, a verdict MUST be written through the MOST RESTRICTIVE separation genuinely available on the host, and the ACHIEVED separation (not the aspired-to one) MUST be reported. **Measured, honestly-degrading limit (spec.md:122):** *"on a single-uid host, capability separation cannot be established by any mechanism an agent can set up — ownership defeats access-control lists, the append-only attribute requires privilege the agent lacks, and any key the producer can use it can also read. Claiming capability there would be the theatre this feature exists to prevent."* A single-uid host therefore reports `instance` honestly, NEVER `capability`, until the operator takes the one action that establishes a real host boundary. **(4) Tier-C trigger 2 — the three named high-blast-radius seams.** Per spec.md:123 (FR-034): at the `release-tag`, `qa-deploy`, or `manual-qa-handoff` seam (or the consuming project's declared equivalent seams per §11.4.35), a producer-authored verdict is REFUSED regardless of the write-access analysis, and the achieved separation is reported rather than assumed. **Same honest-degradation limit applies (spec.md:123):** *"Capability-strength at these three seams requires one operator action (a host boundary agents are correctly forbidden from creating); until it is taken they operate at instance tier and MUST say so."* **(5) No demand for capability outside the two triggers.** Per spec.md:125 (FR-036): at a seam matching NEITHER Tier-C trigger, a separate agent instance MUST SUFFICE and capability separation MUST NOT be demanded — instance-tier verdicts are accepted at ordinary seams without a capability check, closing the negative half of this clause (a project over-applying Tier-C strength to every seam is itself a §11.4.240 violation of the same "false-positive refusal" class). **Honest boundary (§11.4.6).** The `independence_tier` field records WHAT WAS ACHIEVED, never what was ASPIRED to — a project honestly reporting `instance` at a Tier-C-triggered seam because no host boundary yet exists is COMPLIANT with this clause (the honest-degradation path is the compliant path); a project reporting `capability` without the mechanically-verified boundary that makes it true is the

violation. Composes §11.4.6 (honest reporting over aspiration) / §11.4.201 (a false-positive `capability` claim is the same class of bluff a false-positive guard refusal is) / §11.4.256 (its own, distinct, domain-specific different-model requirement stays unweakened) / §11.4.209 (its own, distinct, domain-specific `Fable-xhigh` code-review pin stays unweakened alongside §11.4.256's) / §12 (host-boundary establishment is a §12 host-safety-adjacent operator action, never an autonomous agent action). Recommended mechanism gate

`CM-INDEPENDENCE-TIER-HONESTLY-RECORDED` (every accepted verdict's `independence_tier` field is present, within the closed {instance, model, capability} set, and — at a Tier-C-triggered seam without a mechanically-verified host boundary — is NEVER `capability`; a same-model independent verdict at an ordinary seam is accepted without a capability check) + paired §1.1 mutation (record `capability` at a seam where no verified host boundary exists → the gate MUST FAIL; refuse a same-model independent verdict at an ordinary seam → the gate MUST FAIL [the false-positive-guard case]; accept a producer-authored verdict at `release-tag` / `qa-deploy` / `manual-qa-handoff` → the gate MUST FAIL; golden-FALSE per §11.4.201(1): an `instance`-tier verdict honestly reported at a Tier-C-triggered seam with no host boundary yet established → the gate MUST NOT fire). Gate-code = separate work item, NOT claimed shipped (§11.4.6 / §11.4.227).

Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-240-PROPAGATION` (literal `11.4.240`) + recommended gates `CM-PRODUCER-NOT-VERIFIER-SCOPE-SPLIT` (a change's four scopes per (A) — producer, oracle, gate, verifier — resolve to distinct actors; a producer-authored verdict, a producer-modifiable gate config, a producer-modifiable oracle, or a producer-modifiable test file in the same change violates the split) + `CM-LEAST-PRIVILEGE-CAPABILITY-NOT-INSTRUCTION` (the enforcement channel is a filesystem-scope / branch-protection / verdict-store-isolation / execution-isolation mechanism per (B), NOT an instruction / prompt / policy-document / social-contract; a project whose only enforcement is a written policy that "producers do not touch tests" fails the capability check) + paired §1.1 mutations (make the producer branch a modification to a `tests/*` file within the same change without a non-producer approver → the CODEOWNERS + capability check FAILs; author a status write of "verified: true" from the same actor that produced the diff, with no harness-written §11.4.115(F) verdict pair → the §11.4.146(D3) custody chain FAILs; grant the producer write access to the gate configuration + expect the review to catch the collapse → the

capability-not-instruction gate FAILs on the enforcement-channel check; strip the literal → propagation FAILs; gate-code = separate work item, NOT claimed shipped §11.4.6/§11.4.227).

Canonical authority: constitution submodule Constitution.md §11.4.240. Non-compliance is a release blocker. No escape hatch — no --producer-may-write-tests , --self-certification-OK , --exhortation-suffices-as-enforcement , --producer-authored-verdict-accepted , --capability-limit-optional , --policy-document-is-enforcement , --single-actor-for-all-scopes flag.

§²⁶ 08 15 . . — Illegal-state-unrepresentability preference: types before API-shape before lint before property-test before runtime-assertion (research-derived, — —)

Full anchor text per Phase 3 landing brief. See

curriculum_anchors_241_254_full_drafts.md §11.4.241 canonical block (Part A). Anchor establishes the closed ordered ladder (1) TYPE SYSTEM → (2) API SHAPE → (3) LINT → (4) PROPERTY-BASED TEST → (5) RUNTIME ASSERTION for every load-bearing invariant; a defect preventable at rung N caught only at rung > N is a finding. Composes §11.4.108 (pushes enforcement LEFT of Layer 1) / §11.4.115 / §11.4.142 / §11.4.194 / §11.4.201 / §11.4.224 / §1.1 / §11.4.6 / §11.4.35. Includes rung-by-language capability baseline (strongly-typed: rungs 1-5; dynamically-typed with gradual: rung 1 conditional; untyped scripting: rungs 3-5 unconditional); genuinely-limited-environment clarification per §11.4.209 reviewer MINOR-4 (POSIX sh with only rung 5 SATISFIES the anchor IFF the runtime assertion GENUINELY CATCHES the invariant AND is paired with its §1.1 meta-test mutation per §11.4.107(10) self-validation). Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-241-PROPAGATION (literal 11.4.241) + recommended gate CM-ILLEGAL-STATE-LADDER-PREFERENCE + paired §1.1 mutation (gate-code = separate work item, NOT claimed shipped §11.4.6/§11.4.227).

Canonical authority: constitution submodule `Constitution.md` §11.4.241. Non-compliance is a release blocker. No escape hatch — no `--runtime-assert-suffices`, `--skip-type-preference`, `--dynamic-language-exempts-all-runs`, `--property-test-in-lieu-of-type`, `--decorative-type-annotation-OK` flag.

2026-08-15

§ . . — Byte-identical-fork prohibition + role-as-data-pack (research-derived, — —)

Full anchor per Phase 3 landing brief §11.4.251. When project needs would be met by TWO NEAR-IDENTICAL CODE COPIES differing only in configuration/data/role-specific parameters (prod + staging with hardcoded hostnames; reader + writer with different flag values; web + mobile with different feature flags), copies MUST NOT be maintained as parallel forks. Instead ONE code copy retained + role-specific values extracted into DATA PACK (role-config file / env var set / capability descriptor / feature-flag bundle) consumed by the one code to instantiate as either role. Extraction line 6490 [offset un-resolvable per the §11.4.239 CITATION NOTE; **MODULE ADDED 2026-08-20** — resolves to **module 35** taxonomy row 8 "Byte-identical-fork", verified locators `module_35:127` (row header), `module_35:130` (the SonarQube-Factory case), `module_35:132` (the "roles become data-only packs you reference, never repos you fork" remedy)]: byte-identical-fork taxonomy row + SonarQube-Factory case (`Application/main.kt` byte-for-byte the caching-proxy launcher, still in `package net.milosvasic.factory.proxy.caching.application` — "no SonarQube code"; fix = role-real acceptance test, "roles become data-only packs you reference, never repos you fork"). DETECTION: reviewer or mechanical scan finds byte-identical fork when `diff` shows every difference is a config value (hostname/port/flag/label/env/credential-ref) NOT behavioral, common git-history ancestor, > 90% byte identity; per §11.4.35 threshold-DATA. EXTRACTION: (1) IDENTIFY configuration surface; (2) EXTRACT to data pack per role; (3) REFACTOR single code copy to consume at instantiation (constructor injection / factory / config-loader-at-startup); (4) DELETE one fork; (5) VERIFY per §11.4.5 both role instantiations produce same observable behavior their forks previously produced (§11.4.243 characterization before refactor); (6) commit extraction per §11.4.124 with git-

history evidence. LEGITIMATE FORKS: temporary migration (tracked §11.4.197 end date), experimental / A-B (bounded end date), truly divergent behavior (copies differ in algorithm/protocol/data-model not just config). Composes §11.4.5 / §11.4.6 / §11.4.28 (decoupling — role-configurable code shared across consumers) / §11.4.35 (data-pack location) / §11.4.124 / §11.4.177 / §11.4.194 / §11.4.197 / §11.4.243 / §1.1. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-251-PROPAGATION` (literal `11.4.251`) + recommended gate `CM-NO-BYTE-IDENTICAL-FORK` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.251. Non-compliance is a release blocker. No escape hatch — no `--fork-for-role-OK`, `--skip-data-pack-extraction`, `--copy-and-modify-config`, `--duplicate-code-per-environment`, `--near-identical-forks-are-fine` flag.
